

Computer Networks — Unit 2 Complete README

Data Link Layer (DLL) + Numericals (Hinglish)

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1. Introduction to Data Link Layer

Data Link Layer OSI model ki **2nd layer** hoti hai.

Main Work

Physical layer raw bits bhejti hai.

DLL ensure karti hai:

- Reliable communication
 - Error checking
 - Frame creation
 - Flow control
 - Medium access control
-

2. Functions of Data Link Layer

Function	Meaning
Framing	Data ko frames me convert karna
Physical Addressing	MAC address use karna
Error Control	Error detect/correct
Flow Control	Sender speed control
Access Control	Multiple users manage

3. Design Issues in DLL

Important issues:

1. Framing
2. Error Control
3. Flow Control
4. Access Control

4. Framing

Network layer se data aata hai.

DLL usko small blocks me totti hai called:

Frames

Types of Framing

(A) Character Count

First field batata hai kitne characters hain.

Example:

5HELLO

Problem: Agar count corrupt ho gaya → synchronization lost.

(B) Byte Stuffing

Special characters:

- FLAG
- ESC

Data me FLAG aaye to ESC insert karte hain.

Example:

Before stuffing: FLAG A FLAG B FLAG

After stuffing: FLAG A ESC FLAG B FLAG

(C) Bit Stuffing

Rule: Har 5 consecutive 1s ke baad 0 insert.

Example:

Original: 11111011111

Stuffed: 1111100111110

Numerical — Bit Stuffing

Question

Perform bit stuffing on:

011111101111110

Solution

5 consecutive 1s ke baad 0 add.

Result:

01111101011111010

5. Error Detection & Correction

Transmission me noise se bits change ho sakte hain.

Error Detection Methods

(A) Parity Check

Extra parity bit add hoti hai.

Even Parity

Total 1s even hone chahiye.

Example:

Data: 101101

1s = 4

Parity bit = 0

Final: 1011010

Odd Parity

Total 1s odd hone chahiye.

Numerical — Parity

Question

Find even parity bit for:

1010111

Solution

Number of 1s = 5 (odd)

Need even.

Parity bit = 1

Answer: 10101111

(B) Checksum

Data blocks ko add karte hain.

Complement bhejte hain.

Receiver dubara add karta hai.

Numerical — Checksum

Question

Add: 1010 1100

Solution

$1010 + 1100 = 10110$

Carry wrap: $0110 + 1 = 0111$

Checksum: 1000

(C) CRC (Cyclic Redundancy Check)

Most important.

Polynomial division use hoti hai.

CRC Steps

1. Append zeros
 2. Divide by generator polynomial
 3. Remainder attach
-

Numerical — CRC

Question

Data: 1101011011

Generator: 10011

Find CRC.

Solution

Generator length = 5

Append 4 zeros:

11010110110000

Binary division perform.

Remainder: 1110

Final transmitted data: 11010110111110

Hamming Distance

Minimum bit changes needed.

Formula:

$$2^r \geq m + r + 1$$

Where:

- r = redundant bits
 - m = data bits
-

Numerical — Hamming Code

Question

Find redundant bits for 4-bit data.

Solution

$$m = 4$$

$$\text{Try } r = 3$$

$$2^3 \geq 4 + 3 + 1$$

$$8 \geq 8 \checkmark$$

Answer: 3 redundant bits

6. Flow Control

Fast sender slow receiver ko overload kar sakta hai.

Flow control prevent karta hai.

Types

1. Stop and Wait
 2. Sliding Window
-

Stop and Wait Protocol

Sender:

- ek frame bhejta hai
- ACK wait karta hai

Simple but slow.

Numerical — Stop & Wait Efficiency

Formula:

$$\text{Efficiency} = 1 / (1 + 2a)$$

Where:

- $a = T_p/T_t$

Question

If: $T_p = 20 \text{ ms}$

$T_t = 10 \text{ ms}$

Find efficiency.

Solution

$$a = 20/10 = 2$$

$$\text{Efficiency} = 1 / (1 + 2 \times 2)$$

$$= 1/5$$

$$= 0.2$$

$$= 20\%$$

Sliding Window Protocol

Multiple frames ek saath send hote hain.

Efficient protocol.

Go Back N

Error aaya: $\rightarrow$ uske baad sab resend.

Selective Repeat

Sirf lost frame resend.

More efficient.

Numerical — Sliding Window

Question

If window size = 4

How many frames can sender send without ACK?

Solution

Answer: 4 frames

7. Medium Access Control (MAC)

Same medium multiple users share karte hain.

Collision avoid karna hota hai.

Types of MAC

Type	Protocol
Random Access	ALOHA, CSMA
Controlled Access	Polling
Channelization	FDMA

8. Random Access Protocols

Pure ALOHA

Anytime send.

Efficiency:

$$S = G e^{-2G}$$

Maximum: 18.4%

Numerical — Pure ALOHA

Question

Find throughput when $G = 1$

Solution

$$S = 1 \times e^{-2}$$

$$\approx 0.135$$

$$\approx 13.5\%$$

Slotted ALOHA

Transmission slots me hota hai.

Efficiency:

$$S = Ge^{-G}$$

Maximum: 36.8%

Numerical — Slotted ALOHA

Question

Find throughput for $G = 1$

Solution

$$S = 1 \times e^{-1}$$

$$\approx 0.368$$

$$\approx 36.8\%$$

CSMA

Listen before transmit.

Types

Type	Meaning
1-persistent	Immediately send
Non-persistent	Wait randomly
p-persistent	Send with probability p

CSMA/CD

Collision Detection.

Used in Ethernet.

CSMA/CA

Collision Avoidance.

Used in WiFi.

Difference — CSMA/CD vs CSMA/CA

CSMA/CD	CSMA/CA
Detect collision	Avoid collision
Wired	Wireless
Ethernet	WiFi

9. Controlled Access

Polling

Central controller permission deta hai.

Reservation

Time reserve hota hai.

Token Passing

Token holder hi send karega.

Collision nearly zero.

10. Channelization

FDMA

Different frequencies.

TDMA

Different time slots.

CDMA

Different codes.

Numerical — FDMA

Question

Total bandwidth = 20 MHz

Users = 4

Bandwidth per user?

Solution

$20/4 = 5$ MHz

Answer: 5 MHz

Numerical — TDMA

Question

Frame time = 8 ms

Users = 4

Slot time?

Solution

$8/4 = 2$ ms

Answer: 2 ms

11. Ethernet

IEEE 802.3

Most common LAN.

Types

Ethernet	Speed
Traditional	10 Mbps
Fast	100 Mbps
Gigabit	1 Gbps

Ethernet Frame

Field	Purpose
Preamble	Synchronization
Destination	Receiver MAC
Source	Sender MAC
Type	Protocol

Field	Purpose
Data	Actual data
CRC	Error detection

12. Wireless LAN (IEEE 802.11)

WiFi standard.

Uses:

- CSMA/CA

Advantages:

- mobility
- easy installation

Bluetooth

Short range wireless technology.

Range: ≈ 10 m

Uses:

- headphones
- file sharing
- keyboard

Important Formulas

Stop & Wait Efficiency

$$\text{Efficiency} = 1 / (1 + 2a)$$

Pure ALOHA

$$S = G e^{-2G}$$

Slotted ALOHA

$$S = G e^{(-G)}$$

Hamming Code

$$2^r \geq m+r+1$$

Important Exam Questions

Theory

1. Explain DLL functions
 2. Framing methods
 3. CRC working
 4. Flow control
 5. Sliding window
 6. Ethernet frame
 7. CSMA/CD vs CSMA/CA
 8. Token passing
 9. FDMA vs TDMA vs CDMA
-

Detailed Important Numericals

1. Bit Stuffing Numerical

Question

Perform bit stuffing on:

111110111111011110

Solution

Rule: After every 5 consecutive 1s, insert a 0.

Step:

11111 → insert 0

1111101111110111110

= 11111001111110111110

Again: 11111 → insert 0

Final Answer:

111110011111001111100

2. Even Parity Numerical

Question

Find the even parity bit for:

1011011

Solution

Count number of 1s.

1011011

Number of 1s = 5

Even parity me total 1s even hone chahiye.

So parity bit = 1

Final data:

10110111

3. Odd Parity Numerical

Question

Find odd parity bit for:

1100101

Solution

Number of 1s = 4

Odd parity ke liye total odd hone chahiye.

Parity bit = 1

Final Answer:

11001011

4. Checksum Numerical

Question

Find checksum of:

10101010 11001100

Solution

Binary addition:

10101010 + 11001100

101110110

Carry wrap:

01110110 + 1

= 01110111

1's complement:

10001000

Final checksum:

10001000

5. CRC Numerical

Question

Data = 1010001101

Generator = 1101

Find CRC.

Solution

Step 1: Generator bits = 4

Append 3 zeros:

1010001101000

Step 2: Perform modulo-2 division.

Remainder obtained:

011

Step 3: Append remainder to original data.

Final transmitted data:

1010001101011

6. Hamming Code Numerical

Question

Find number of redundant bits for 7-bit data.

Solution

Formula:

$$2^r \geq m + r + 1$$

Where: $m = 7$

Try $r = 4$

$$2^4 = 16$$

$$m + r + 1 = 7 + 4 + 1 = 12$$

$$16 \geq 12 \checkmark$$

Answer: 4 redundant bits

7. Stop and Wait Efficiency Numerical

Question

Propagation time = 30 ms

Transmission time = 10 ms

Find efficiency.

Solution

Formula:

$$\text{Efficiency} = 1 / (1 + 2a)$$

Where:

$$a = T_p / T_t$$

$$= 30 / 10$$

$$= 3$$

Efficiency:

$$= 1 / (1 + 2 \times 3)$$

$$= 1/7$$

$$= 0.142$$

$$= 14.2\%$$

8. Sliding Window Numerical

Question

If sender window size = 5, find maximum frames that can be sent before ACK.

Solution

In sliding window, sender can send frames equal to window size.

Answer:

5 frames

9. Pure ALOHA Numerical

Question

Find throughput when:

$G = 0.5$

Solution

Formula:

$$S = G e^{-2G}$$

Put $G = 0.5$

$$S = 0.5 \times e^{-1}$$

$$= 0.5 \times 0.368$$

$$= 0.184$$

$$= 18.4\%$$

10. Slotted ALOHA Numerical

Question

Find throughput for:

$$G = 1$$

Solution

Formula:

$$S = G e^{(-G)}$$

$$= 1 \times e^{(-1)}$$

$$= 0.368$$

$$= 36.8\%$$

11. FDMA Numerical

Question

Total bandwidth = 40 MHz

Users = 8

Bandwidth per user?

Solution

Bandwidth per user:

$$= 40 / 8$$

$$= 5 \text{ MHz}$$

Answer: 5 MHz

12. TDMA Numerical

Question

Total frame time = 12 ms

Users = 6

Find slot time.

Solution

Slot time:

$$= 12 / 6$$

$$= 2 \text{ ms}$$

Answer: 2 ms

13. Transmission Delay Numerical

Question

Frame size = 2000 bits

Bandwidth = 2 Mbps

Find transmission delay.

Solution

Formula:

$$\text{Transmission Delay} = \text{Frame Size} / \text{Bandwidth}$$

$$= 2000 / (2 \times 10^6)$$

$$= 0.001 \text{ sec}$$

$$= 1 \text{ ms}$$

14. Propagation Delay Numerical

Question

Distance = 3000 km

Propagation speed = 3×10^8 m/s

Find propagation delay.

Solution

Convert distance:

3000 km = 3×10^6 m

Formula:

Propagation Delay = Distance / Speed

= $(3 \times 10^6) / (3 \times 10^8)$

= 10^{-2} sec

= 0.01 sec

= 10 ms

Viva Questions

Question	Answer
Which layer handles framing?	DLL
Which protocol uses token?	Token passing
CSMA/CD used where?	Ethernet
CSMA/CA used where?	WiFi
IEEE standard for Ethernet?	802.3
IEEE standard for WiFi?	802.11

One-Line Revision

Topic	Quick Revision
DLL	Reliable delivery
Framing	Data division
CRC	Strong detection
Hamming	Error correction
Stop & Wait	One frame at time
Sliding Window	Multiple frames
Ethernet	Wired LAN
WiFi	Wireless LAN

Detailed Theory Notes

Framing Detailed Theory

Framing ka main purpose data ko manageable units me divide karna hota hai.

Receiver ko pata chalna chahiye:

- frame kaha start hua
- frame kaha end hua

Agar framing na ho:

- receiver confuse ho jayega
- synchronization lost ho sakti hai

Types of Framing

1. Character Count

Frame ke start me count diya jata hai.

Example:

5HELLO

Yaha 5 batata hai ki total 5 characters hain.

Advantage

- Simple

Disadvantage

- Count corrupt ho jaye to pura frame corrupt.
-

2. Byte Stuffing

Special bytes use hoti hain:

- FLAG
- ESC

Agar data me FLAG aa jaye to ESC insert karte hain.

Advantage

- Better synchronization

Disadvantage

- Extra bytes increase overhead.
-

3. Bit Stuffing

Rule: Har 5 consecutive 1s ke baad 0 insert.

Mostly HDLC protocol me use hota hai.

Advantage

- Efficient
 - Reliable
-

Error Detection Detailed Theory

Transmission medium me noise hota hai.

Noise ki wajah se:

- bits flip ho sakti hain
- data corrupt ho sakta hai

DLL error detect aur correct karti hai.

Parity Check

Extra bit add karte hain.

Even Parity

Total 1s even hone chahiye.

Odd Parity

Total 1s odd hone chahiye.

Advantages

- Simple
- Fast

Disadvantages

- Multiple bit errors detect nahi karta.
-

CRC Detailed Theory

CRC sabse powerful error detection technique hai.

Modulo-2 division use hoti hai.

Sender:

- zeros append karta hai
- divide karta hai
- remainder attach karta hai

Receiver:

- dubara divide karta hai

Remainder 0: → no error

Remainder non-zero: → error detected

Advantages

- Strong error detection
 - Burst errors detect karta hai
-

Hamming Code Detailed Theory

Hamming code single-bit error:

- detect bhi
- correct bhi

kar sakta hai.

Redundant bits powers of 2 positions par rakhe jate hain.

Positions: 1, 2, 4, 8...

Flow Control Detailed Theory

Fast sender slow receiver ko overload kar sakta hai.

Isliye flow control use hota hai.

Stop and Wait Protocol

Sender: 1. Frame bhejta hai 2. ACK wait karta hai 3. ACK aaye to next frame bhejta hai

Advantages

- Simple
- Easy implementation

Disadvantages

- Low efficiency
 - Slow communication
-

Sliding Window Protocol

Multiple frames ek saath bhej sakte hain.

Window size define hoti hai.

Advantages

- High efficiency
- Better bandwidth utilization

Disadvantages

- Complex implementation
-

Go Back N ARQ

Agar ek frame lost ho jaye: → uske baad wale sab frames resend.

Advantage

- Simple receiver design

Disadvantage

- Bandwidth waste hota hai.
-

Selective Repeat ARQ

Sirf lost frame resend hota hai.

Advantage

- Efficient

Disadvantage

- Complex buffering required.
-

MAC Detailed Theory

MAC ka full form: Medium Access Control

Same channel multiple devices share karti hain.

Collision avoid karne ke liye MAC protocols use hote hain.

Pure ALOHA

Station jab chahe transmit kare.

Collision chances high.

Maximum efficiency: 18.4%

Slotted ALOHA

Transmission fixed slots me hota hai.

Collision kam hota hai.

Maximum efficiency: 36.8%

CSMA

Carrier Sense Multiple Access

Rule: Listen before transmit.

Working

- Channel free → send
 - Busy → wait
-

CSMA/CD

Collision Detection.

Used in Ethernet.

Steps: 1. Listen 2. Send 3. Detect collision 4. Stop transmission 5. Wait random time 6. Retransmit

CSMA/CA

Collision Avoidance.

Used in WiFi.

Wireless me collision detect karna difficult hota hai.

Isliye avoid kiya jata hai.

Controlled Access Detailed Theory

Polling

Central controller har device se poochta hai.

Advantage

- No collision

Disadvantage

- Slow
-

Reservation

Transmission se pehle reservation.

Advantage

- Organized communication
-

Token Passing

Ek token circulate karta hai.

Jiske paas token: → wahi send karega.

Advantage

- Collision almost zero

Disadvantage

- Token loss issue
-

Channelization Detailed Theory

FDMA

Har user ko separate frequency.

Used in radio systems.

TDMA

Har user ko separate time slot.

Used in GSM.

CDMA

Sab same frequency use karte hain but unique codes.

Advantages

- Secure
 - Efficient
-

Ethernet Detailed Theory

IEEE 802.3 standard.

Most common wired LAN.

Ethernet Types

Type	Speed
Traditional Ethernet	10 Mbps
Fast Ethernet	100 Mbps
Gigabit Ethernet	1 Gbps
10 Gigabit Ethernet	10 Gbps

Ethernet Frame Detailed Fields

Field	Size	Purpose
Preamble	7 bytes	Synchronization
Destination Address	6 bytes	Receiver MAC
Source Address	6 bytes	Sender MAC
Type	2 bytes	Protocol type
Data	46–1500 bytes	Actual data

Field	Size	Purpose
CRC	4 bytes	Error checking

Wireless LAN Detailed Theory

IEEE 802.11 standard.

Features:

- wireless communication
- mobility
- easy installation
- flexible network

Uses:

- homes
- offices
- colleges

Bluetooth Detailed Theory

Short-range wireless communication technology.

Range: $\approx$ 10 meters

Uses:

- earbuds
- keyboards
- speakers
- file transfer

Important Tips for Exams

Learn These Properly

1. CRC numerical
2. Hamming code
3. Stop & Wait efficiency
4. ALOHA throughput
5. CSMA/CD vs CSMA/CA
6. Ethernet frame

Mind Map — Unit 2 Data Link Layer



